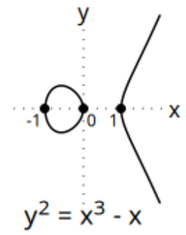
Fermat's last theorem for n=4
(any field K has been replaced with $\mathbb{Q}$)
from: Kato, Kazuya, Nobushige Kurokawa, and Takeshi Saitō.
Number Theory: Fermat's dream. AMS Bookstore, 2000.

Def * Elliptic curve over $\mathbb{Q}$ is equation of form:
$y^2 = ax^3 + bx^2 + cx + d$ $(a, b, c, d \in Q, a \neq 0)$
where RHS polynomial has no multiple root

Def For $a = \frac{m}{n}$ with $m, n \in \mathbb{N},\ (m, n) = 1$
define height $H(a) := max(|m|, |n|)$

(x, y) on elliptic curve
$\implies$ $(\frac{-1}{x}, \frac{y}{x^2})$ on curve as well,
$H(x) = H(-1/x)$

Lem 1.5 For distinct $a, b, c \in \mathbb{Q}$ define $B, \tilde{C}$ and C
$B = \{(u, v, w) \in \mathbb{Q} \times \mathbb{Q} \times \mathbb{Q} \mid u^2 + a = v^2 + b = w^2 + c\}$
$\tilde{C} = \{(x, y) \in \mathbb{Q} \times \mathbb{Q} \mid y^2 = (x - a)(x - b)(x - c)\}$.
$C = \{(x, y) \in \mathbb{Q} \times \mathbb{Q} \mid y^2 = (x - a)(x - b)(x - c),\ y \neq 0\}$.
(1) $\exists$ mutually inverse maps $f : B \to C,\ g : C \to B$
$f(u, v, w) = (u^2 + a + uv + vw + wu,\ (u + v)(v + w)(w + u))$,
$g(x, y) = \Big(\frac{1}{2y}((x - a)^2 - (b - a)(c - a)),$
$\frac{1}{2y}((x - b)^2 - (a - b)(c - b)),\ \frac{1}{2y}((x - c)^2 - (a - c)(b - c))\Big)$
(2) $\exists$ map $h : B \to \tilde{C},\ h(u, v, w) = (u^2 + a, uvw)$



(i) Let (x_0, y_0) on $y^2 = x^3 - x$ with $y_0 \neq 0$
and minimal $H(x_0)$; by above we can assume $1 < x_0$

$i : (x, y, z) \mapsto \left(\frac{y-x}{2}, \frac{z}{2}, \frac{x+y}{2}\right)$
$j : (u, v, w) \mapsto (w - u, w + u, 2v)$

$\begin{cases} i : A_d \to B_d \\ j : B_d \to A_d \end{cases}$

$a = d,\ b = 0,\ c = -d,$ $\begin{cases} f : B_d \to C_d \\ g : C_d \to B_d \end{cases}$

Rem 1.6 $h \circ g : C \to \tilde{C}$ is called multiplication-by-2 map
for elliptic curve $y^2 = (x - a)(x - b)(x - c)$; image of $h \circ g$ is
$\{(x, y) \in \mathbb{Q} \times \mathbb{Q} \mid y^2 = (x - a)(x - b)(x - c),\ x - a,\ x - b,\ x - c$ are $\square$s in $\mathbb{Q}\}$
because:
$(x', y') = h(u, v, w) \implies x' = u^2 + a \implies x' - a$ is $\square$ in $\mathbb{Q}$
from definition of B (image of g): $u^2 + a = v^2 + b = w^2 + c$
$x' - b = (v^2 + b) - b = v^2$ is $\square$ in $\mathbb{Q}$
$x' - c = (w^2 + c) - c = w^2$ is $\square$ in $\mathbb{Q}$

a=1, b=0, c=-1

Lem 1.4 For $d \in \mathbb{Q}$ define:
$A_d = \{(x, y, z) \in \mathbb{Q} \times \mathbb{Q} \times \mathbb{Q} \mid x^2 + y^2 = z^2, xy = d\}$
$B_d = \{(u, v, w) \in \mathbb{Q} \times \mathbb{Q} \times \mathbb{Q} \mid u^2 + d = v^2,\ v^2 + d = w^2\}$
$C_d = \{(x, y) \in \mathbb{Q} \times \mathbb{Q} \mid y^2 = x^3 - d^2x,\ y \neq 0\}$.
$\exists$ bijections between any 2 of A_d, B_d, C_d

$\begin{cases} (i) & \implies A_d \neq \emptyset \\ (ii) & \implies B_d \neq \emptyset \\ (iii) & \implies C_d \neq \emptyset \end{cases}$

Prop 0.13 The area of a right $\triangle$
whose sides are integers is not a $\square$

$\neg(i)$
$d = 1$

Lem 1.3 $d \in \mathbb{Q}$, these are equivalent:
(i) $\exists$ rational number sides **right** $\triangle$ of area d
(ii) $\exists$ 3 $\square$s of rational numbers that form
arithmetic progression of difference d
(iii) $\exists$ rational solution to $y^2 = x^3 - d^2x$
other than $(x, y) = (0, 0)$ and $(\pm d, 0)$

$\neg(iii)$
$d = 1$

Prop 1.2 The only rational solutions to
$y^2 = x^3 - x$ are $(x, y) = (0, 0)$ and $(\pm 1, 0)$

(ii) $x_0 - 1$, x_0 and $x_0 + 1$ are all $\square$s
$\implies$ $\exists (x_1, y_1) \in C : h \circ g(x_1, y_1) = (x_0, y_0)$

multiply a 1.1 solution by $\frac{z^2}{y^6}$, $\lightning$:

$$0 \neq \left(\frac{x^2 z}{y^3}\right)^2 = \left(\frac{z^2}{y^2}\right)^3 - \frac{z^2}{y^2}$$

Prop 1.1 $\nexists$ solution (x, y, z) to
$x^4 + y^4 = z^4$ satisfying $xyz \neq 0$

(iii) $H(x_1) < H(x_0)$, $\lightning$